SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 06 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I **V.Madhulika(192312620)***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

v. madhulika

Signature of the student:

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

Answer:

A typical machine learning system consists of the following functional blocks arranged as a pipeline:

- *Data Collection – gathering raw data from sensors, databases, logs, or the web.*
- *Data Pre-processing – cleaning missing/noisy values, normalising, encoding categorical data.*
- *Feature Extraction/Selection – choosing the most relevant attributes that influence the output.*
- *Model Selection – choosing an algorithm (e.g., regression, decision tree, neural network) suited to the problem.*
- *Training – the model learns parameters/patterns from the training dataset.*
- *Evaluation/Validation – testing the model on unseen data using metrics such as accuracy, precision, recall.*
- *Deployment & Feedback – the model is deployed for prediction and continuously improved using new data (feedback loop).*

Supervised Learning Example:

Email spam detection – the model is trained on emails that are already labelled as "Spam" or "Not Spam". It learns the mapping between input features (words, sender, links) and the known output label, and then classifies new, unseen emails.

Unsupervised Learning Example:

Customer segmentation – a retail company groups customers based on purchasing behaviour without any predefined labels. The algorithm (e.g., K-Means) discovers natural clusters such as "budget shoppers" and "premium shoppers" purely from the structure of the data.

2. Design a machine learning system for disaster management to predict events such as floods and earthquakes using historical and real-time data. Explain the choice of model, data preprocessing, and feature selection techniques. (a) Describe the steps involved in training, validating, and evaluating the model. (b) Explain how the system can assist in early warning and emergency response. (5 Marks)

Answer:

Model Choice:

A hybrid approach works best: Random Forest / XGBoost for structured tabular sensor data (rainfall, river level, seismic magnitude), and an LSTM (Long Short-Term Memory) network for time-series data such as continuous rainfall or seismic readings, since disaster precursors evolve over time.

Data Pre-processing:

- *Handling missing sensor readings using interpolation or forward-fill.*
- *Normalising values (rainfall in mm, seismic magnitude, water level) to a common scale.*
- *Removing outliers/noise caused by faulty sensors.*
- *Merging historical records with real-time IoT/satellite feeds into a unified timestamped dataset.*

Feature Selection:

Key features include rainfall intensity, river discharge level, soil moisture, seismic frequency/magnitude, tectonic plate movement data, and historical disaster occurrence in the region. Techniques such as correlation analysis and Recursive Feature Elimination (RFE) are used to retain only the most predictive features.

(a) Training, Validation and Evaluation:

- *Split data into training (70%), validation (15%) and test (15%) sets, preserving time order for time-series data.*
- *Train the chosen model on historical labelled disaster/no-disaster events.*
- *Use k-fold or time-series cross-validation to tune hyperparameters and avoid overfitting.*
- *Evaluate using recall and F1-score (favoured over plain accuracy since disaster events are rare/imbalanced), along with ROC-AUC.*

(b) Role in Early Warning and Emergency Response:

The trained model continuously scores incoming real-time data; when the predicted probability of a flood/earthquake crosses a threshold, the system automatically triggers SMS/siren alerts to residents and authorities, recommends evacuation routes, and shares risk-zone maps with emergency responders — reducing response time and potential loss of life.

3. In the context of Find-S algorithm, consider the following dataset used to determine whether a day is suitable for playing tennis based on attributes such as Outlook, Temperature, Humidity, and Wind. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show the step-by-step updates of the hypothesis after each positive training example and ignore all negative examples. Finally, provide the final hypothesis. (5 Marks)

Answer:

Only positive examples (Play Tennis = Yes) are considered: Examples 3, 4, 5 and 7.

Initial hypothesis: $h_0 = \langle \varnothing, \varnothing, \varnothing, \varnothing \rangle$

- *Example 3 (Overcast, Hot, High, Weak, Yes):* $h_1 = \langle Overcast, Hot, High, Weak \rangle$
- *Example 4 (Rainy, Mild, High, Weak, Yes): Outlook and Temperature differ from* h_1 *→ generalise to '?'.* $h_2 = \langle ?, ?, High, Weak \rangle$
- *Example 5 (Rainy, Cool, Normal, Weak, Yes): Humidity differs (High vs Normal) → generalise.* $h_3 = \langle ?, ?, ?, Weak \rangle$
- *Example 7 (Overcast, Mild, Normal, Strong, Yes): Wind differs (Weak vs Strong) → generalise.* $h_4 = \langle ?, ?, ?, ? \rangle$

Final Hypothesis: $h = \langle ?, ?, ?, ? \rangle$

This means that, based purely on the positive examples given, Find-S is unable to find any restrictive pattern common to all of them, so it generalises to "any day is suitable for tennis" — illustrating a limitation of Find-S: it ignores negative examples and can over-generalise when positive examples are diverse.

4. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a person will enjoy a sport based on attributes such as Weather, Temperature, Humidity, and Wind. Using the Candidate Elimination algorithm, determine the version space by computing both the specific (S) and general (G) boundaries. Show the step-by-step updates after processing each training example. (5 Marks)

Answer:

Initial boundaries: $S_0 = \langle \varnothing, \varnothing, \varnothing, \varnothing \rangle$, $G_0 = \langle ?, ?, ?, ? \rangle$

- *Example 1 (Sunny, Warm, Normal, Weak, Yes) — positive:* $S_1 = \langle Sunny, Warm, Normal, Weak \rangle$; $G_1 = \langle ?, ?, ?, ? \rangle$
- *Example 2 (Sunny, Warm, High, Strong, Yes) — positive: Humidity and Wind differ from* S_1*, so generalise:* $S_2 = \langle Sunny, Warm, ?, ? \rangle$; $G_2 = \langle ?, ?, ?, ? \rangle$

- *Example 3 (Rainy, Cold, High, Weak, No) — negative: specialise G to exclude it while staying consistent with S_2 : G_3 = { ⟨Sunny, ?, ?, ?⟩, ⟨?, Warm, ?, ?⟩ }*
- *Example 4 (Sunny, Cold, Normal, Weak, Yes) — positive: Temperature differs from S_2 , generalise: S_3 = ⟨Sunny, ?, ?, ?⟩. Remove ⟨?, Warm, ?, ?⟩ from G (inconsistent, since Temperature = Cold here): G_4 = { ⟨Sunny, ?, ?, ?⟩ }*
- *Example 5 (Rainy, Warm, Normal, Strong, No) — negative: already excluded by G_4 (Weather ≠ Sunny), no change needed: G_5 = { ⟨Sunny, ?, ?, ?⟩ }*

Final Version Space: S = ⟨Sunny, ?, ?, ?⟩ and G = { ⟨Sunny, ?, ?, ?⟩ }. Since S = G, the version space converges to a single hypothesis: "Enjoy Sport = Yes whenever Weather = Sunny", regardless of the other attributes.

5. In the context of the Find-S algorithm, consider the dataset used to determine whether a customer will buy a product based on attributes such as Age, Income, Student, and Credit Rating. Apply the Find-S algorithm to determine the maximally specific hypothesis. Show step-by-step updates after each positive example and ignore negative examples. Finally, provide the final hypothesis. (5 Marks)

Answer:

Only positive examples (Buys Product = Yes) are used: Examples 3 and 4.

Initial hypothesis: h_0 = ⟨∅, ∅, ∅, ∅⟩

- *Example 3 (Middle, High, No, Fair, Yes): h_1 = ⟨Middle, High, No, Fair⟩*
- *Example 4 (Senior, Medium, No, Fair, Yes): Age (Middle→Senior) and Income (High→Medium) differ → generalise to '?'. h_2 = ⟨?, ?, No, Fair⟩*

Final Hypothesis: h = ⟨?, ?, No, Fair⟩ — a customer buys the product whenever they are Not a Student and have a Fair credit rating, irrespective of age or income.

6. In the context of the Find-S algorithm, consider a dataset used to determine whether a person will go for a workout based on attributes such as Weather, Temperature, Time, and Mood. Apply the Find-S algorithm and determine the maximally specific hypothesis. Show the hypothesis updates after each positive example only. (5 Marks)

Answer:

Only positive examples (Workout = Yes) are used: Examples 2, 3 and 4.

Initial hypothesis: h_0 = ⟨∅, ∅, ∅, ∅⟩

- *Example 2 (Rainy, Mild, Evening, Happy, Yes): h_1 = ⟨Rainy, Mild, Evening, Happy⟩*
- *Example 3 (Sunny, Mild, Evening, Happy, Yes): Weather differs (Rainy→Sunny) → generalise. h_2 = ⟨?, Mild, Evening, Happy⟩*
- *Example 4 (Cloudy, Cool, Morning, Happy, Yes): Temperature and Time differ → generalise. h_3 = ⟨?, ?, ?, Happy⟩*

Final Hypothesis: h = ⟨?, ?, ?, Happy⟩ — the person goes for a workout whenever their Mood is Happy, regardless of weather, temperature, or time.

7. In the context of the Find-S algorithm, consider a dataset used to determine whether a loan should be approved based on attributes such as Employment Status, Salary Level, Credit Score, and Loan History. Apply the Find-S algorithm to derive the most specific hypothesis consistent with the positive examples. (5 Marks)

Answer:

Only positive examples (Loan Approved = Yes) are used: Examples 1, 2 and 4.

Initial hypothesis: $h_0 = \langle\varnothing, \varnothing, \varnothing, \varnothing\rangle$

- *Example 1 (Employed, High, Good, Clean, Yes): $h_1 = \langle$Employed, High, Good, Clean$\rangle$*
- *Example 2 (Employed, Medium, Good, Clean, Yes): Salary differs (High→Medium) → generalise. $h_2 = \langle$Employed, ?, Good, Clean$\rangle$*
- *Example 4 (Employed, Medium, Excellent, Clean, Yes): Credit Score differs (Good→Excellent) → generalise. $h_3 = \langle$Employed, ?, ?, Clean$\rangle$*

Final Hypothesis: $h = \langle$Employed, ?, ?, Clean$\rangle$ — a loan is approved whenever the applicant is Employed and has a Clean loan history, irrespective of salary level or credit score.

8. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a customer will subscribe to a service based on attributes such as Age Group, Income Level, Internet Usage, and Device Type. Determine the version space by computing the specific (S) and general (G) boundaries. Show step-by-step updates after each training example. (5 Marks)

Answer:

Initial boundaries: $S_0 = \langle\varnothing, \varnothing, \varnothing, \varnothing\rangle$, $G_0 = \langle ?, ?, ?, ?\rangle$

- *Example 1 (Young, High, Frequent, Mobile, Yes) — positive: $S_1 = \langle$Young, High, Frequent, Mobile$\rangle$; $G_1 = \langle ?, ?, ?, ?\rangle$*
- *Example 2 (Young, High, Occasional, Mobile, Yes) — positive: Internet Usage differs → generalise. $S_2 = \langle$Young, High, ?, Mobile$\rangle$; $G_2 = \langle ?, ?, ?, ?\rangle$*
- *Example 3 (Middle, Low, Frequent, Desktop, No) — negative: specialise G consistent with S_2 : $G_3 = \{ \langle$Young, ?, ?, ?$\rangle$, $\langle$?, High, ?, ?$\rangle$, $\langle$?, ?, ?, Mobile$\rangle \}$*
- *Example 4 (Young, Medium, Frequent, Mobile, Yes) — positive: Income differs from S_2 → generalise. $S_3 = \langle$Young, ?, ?, Mobile$\rangle$. Remove $\langle$?, High, ?, ?$\rangle$ from G (inconsistent, Income = Medium here): $G_4 = \{ \langle$Young, ?, ?, ?$\rangle$, $\langle$?, ?, ?, Mobile$\rangle \}$*
- *Example 5 (Senior, High, Occasional, Desktop, No) — negative: already excluded by both members of G_4 (Age ≠ Young and Device ≠ Mobile), no change: $G_5 = \{ \langle$Young, ?, ?, ?$\rangle$, $\langle$?, ?, ?, Mobile$\rangle \}$*

Final Version Space: S = $\langle$Young, ?, ?, Mobile$\rangle$ and G = $\{ \langle$Young, ?, ?, ?$\rangle$, $\langle$?, ?, ?, Mobile$\rangle \}$.

9. In the context of the Candidate Elimination algorithm, consider a dataset used to classify whether a patient follows a healthy lifestyle based on attributes such as Diet, Exercise, Sleep, and Stress Level. Apply the algorithm to compute the version space (S and G boundaries) and show the updates after each example. (5 Marks)

Answer:

Initial boundaries: $S_0 = \langle\varnothing, \varnothing, \varnothing, \varnothing\rangle$, $G_0 = \langle ?, ?, ?, ?\rangle$

- *Example 1 (Balanced, Regular, Adequate, Low, Yes) — positive: $S_1 = \langle$Balanced, Regular, Adequate, Low$\rangle$; $G_1 = \langle ?, ?, ?, ?\rangle$*
- *Example 2 (Balanced, Regular, Poor, Medium, Yes) — positive: Sleep and Stress differ → generalise. $S_2 = \langle$Balanced, Regular, ?, ?$\rangle$; $G_2 = \langle ?, ?, ?, ?\rangle$*
- *Example 3 (Junk, None, Poor, High, No) — negative: specialise G consistent with S_2 : $G_3 = \{ \langle$Balanced, ?, ?, ?$\rangle$, $\langle$?, Regular, ?, ?$\rangle \}$*
- *Example 4 (Balanced, Moderate, Adequate, Low, Yes) — positive: Exercise differs from S_2 → generalise. $S_3 = \langle$Balanced, ?, ?, ?$\rangle$. Remove $\langle$?, Regular, ?, ?$\rangle$ from G (inconsistent, Exercise = Moderate here): $G_4 = \{ \langle$Balanced, ?, ?, ?$\rangle \}$*

- *Example 5 (Junk, Regular, Adequate, High, No) — negative: already excluded by G_4 (Diet ≠ Balanced), no change: G_5 = { ⟨Balanced, ?, ?, ?⟩ }*

Final Version Space: S = G = { ⟨Balanced, ?, ?, ?⟩ }. The version space converges to a single hypothesis: "Healthy Lifestyle = Yes if and only if Diet = Balanced."

10. Design a machine learning system for health analysis to predict diseases such as heart disease or diabetes using patient data including age, blood pressure, glucose level, and lifestyle factors. Explain the choice of suitable algorithms and justify their selection. Describe the steps involved in data preprocessing, feature selection, and model training to ensure accurate and reliable predictions. (5 Marks)

Answer:

Algorithm Choice:

Logistic Regression is used as a baseline for its interpretability (important in healthcare), while Random Forest / XGBoost are preferred for the final model since they handle non-linear relationships between features (e.g., blood pressure and glucose interactions) and are robust to outliers and missing data, which are common in patient records.

Data Pre-processing:

- *Impute missing clinical values using mean/median or model-based imputation.*
- *Normalise numeric attributes (age, BP, glucose) using StandardScaler so no single feature dominates.*
- *Encode categorical lifestyle factors (smoking, exercise) using one-hot encoding.*
- *Address class imbalance (fewer disease-positive cases) with SMOTE or class-weighting.*

Feature Selection:

Use correlation analysis and feature-importance scores from the trained Random Forest to retain the most predictive attributes (e.g., glucose level and BMI for diabetes; blood pressure and cholesterol for heart disease), discarding redundant ones to reduce overfitting.

Model Training:

Split the data (train/validation/test), apply k-fold cross-validation for hyperparameter tuning, and evaluate using sensitivity (recall), specificity, and ROC-AUC — metrics that matter more than raw accuracy for medical diagnosis, where missing a true positive can be costly.

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

Answer:

Pipeline diagram (linear flow with feedback loop):

Data Collection → Data Pre-processing → Feature Engineering → Model Training → Model Evaluation → Deployment → Monitoring/Feedback → (back to Data Collection)

- *Data Collection: gathering raw data from databases, sensors, APIs, or user interactions.*
- *Pre-processing: cleaning, handling missing values, removing duplicates/outliers.*
- *Feature Engineering: creating, transforming, or selecting attributes that best represent the underlying problem.*
- *Model Training: fitting an algorithm to the training data so it learns the mapping from inputs to outputs.*
- *Evaluation: measuring performance on a held-out test set using appropriate metrics.*
- *Deployment: integrating the trained model into a production system to serve real-time predictions, followed by ongoing monitoring for drift.*

Training Error vs Generalization Error:

Training error is the error the model makes on the data it was trained on, while generalization error is the error on new, unseen data. For example, a very deep decision tree may achieve near-zero training error by memorising every training example, but perform poorly on the test set (high generalization error) — this gap indicates overfitting. A well-generalising model keeps both errors low and close to each other.

12. Design a machine learning model for spam email classification using labeled data. (a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression). (b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization. (c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score. (5 Marks)

Answer:

(a) Algorithm Choice:

Naïve Bayes (specifically Multinomial Naïve Bayes) is well suited for spam detection because it works efficiently with high-dimensional, sparse text data, assumes word-independence (a reasonable approximation for bag-of-words text), trains quickly, and performs well even with relatively small labelled datasets. SVM can be used as an alternative for higher accuracy when more computing power is available, since it handles high-dimensional feature spaces well.

(b) Pre-processing Steps:

- *Tokenization: splitting each email into individual words/tokens.*
- *Stop-word Removal: removing common, non-informative words (e.g., "the", "is").*
- *Normalisation: lower-casing text, removing punctuation, and stemming/lemmatising words.*
- *Vectorization: converting text into numeric form using Bag-of-Words or TF-IDF, which weighs words by their importance/frequency.*

(c) Evaluation Metrics:

- *Accuracy: overall proportion of correctly classified emails.*
- *Precision: of the emails predicted as spam, how many were actually spam — important to avoid marking genuine emails as spam.*
- *Recall: of all actual spam emails, how many were correctly identified.*
- *F1-score: the harmonic mean of precision and recall, giving a balanced measure — especially useful since spam datasets are often imbalanced.*

13. In the context of the Candidate-Elimination Algorithm, consider a dataset with attributes: Sky, AirTemp, Humidity, Wind, Water, Forecast and target concept EnjoySport. (a) Explain the initialization of S (specific boundary) and G (general boundary). (b) Show how S and G change with each training example. (c) Compare Candidate-Elimination with the Find-S algorithm in terms of generalization capability. (5 Marks)

Answer:

(a) Initialization:

The specific boundary S is initialised to the most specific hypothesis possible: $S_0 = \langle \varnothing, \varnothing, \varnothing, \varnothing, \varnothing, \varnothing \rangle$, representing a hypothesis that classifies every instance as negative (no positive instance has been seen yet). The general boundary G is initialised to the most general hypothesis: $G_0 = \langle ?, ?, ?, ?, ?, ? \rangle$, representing a hypothesis that classifies every instance as positive.

(b) Updates with each Training Example:

- *For a positive example: any hypothesis in G that does not cover it is removed, and hypotheses in S that don't cover it are generalised minimally so that they cover the new example while remaining as specific as possible.*

- *For a negative example: any hypothesis in S that covers it is removed, and hypotheses in G are specialised minimally so that they exclude the negative example while remaining as general as possible, and still consistent with S.*
- *This continues example by example until S and G converge (possibly to a single hypothesis) or all training data is processed, giving the final Version Space — the set of all hypotheses consistent with the data.*

(c) Comparison with Find-S:

- *Find-S only tracks the specific boundary S and completely ignores negative examples, so it cannot detect inconsistent/noisy training data and gives only one hypothesis.*
- *Candidate-Elimination maintains both S and G, using both positive and negative examples, and represents the entire space of consistent hypotheses (the Version Space) rather than a single hypothesis.*
- *Consequently, Candidate-Elimination has stronger generalization capability and more expressive power, since it can identify ambiguity (multiple hypotheses still consistent) and detect inconsistencies, while Find-S offers a computationally cheaper but less informative result.*

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest). (b) Explain feature scaling and handling missing data. (c) Describe how overfitting can be detected and avoided. (5 Marks)

Answer:

(a) Model Choice:

Random Forest Regression is generally preferred over plain Linear Regression because house price relationships (location, size, amenities) are often non-linear and involve complex feature interactions that a single linear model cannot capture. Random Forest, being an ensemble of decision trees, handles non-linearity and mixed data types well and is more robust to outliers than a single Decision Tree, which tends to overfit on its own. Linear Regression remains useful as a simple, interpretable baseline.

(b) Feature Scaling and Missing Data:

Feature scaling (e.g., StandardScaler or Min-Max normalisation) is applied so that features on different scales, such as size in square feet versus number of rooms, contribute proportionately during training rather than one dominating due to its larger numeric range — this matters most for distance-based or gradient-based models. Missing data (e.g., unreported amenities) is handled via mean/median/mode imputation for numeric/categorical fields, or by using models like Random Forest that can tolerate some missingness directly.

(c) Overfitting Detection and Avoidance:

- *Detection: a large gap between low training error and high validation/test error, or a learning curve where validation error rises while training error keeps falling.*
- *Avoidance: use k-fold cross-validation, apply regularization (L1/L2 for linear models), prune decision trees, limit tree depth, use ensemble methods (Random Forest) which average out variance, and gather more training data where possible.*

15. Explain the concept of clustering in unsupervised learning. (a) Describe the working of the K-Means algorithm with a step-by-step example. (b) Explain how the value of K is chosen using the Elbow Method. (c) Compare K-Means with Hierarchical Clustering. (5 Marks)

Answer:

Clustering is an unsupervised learning technique that groups unlabelled data points into clusters such that points within the same cluster are more similar to each other than to points in other clusters, based purely on the structure of the data (no target labels are used).

(a) K-Means Algorithm — Step by Step:

- *Step 1: Choose the number of clusters, K, and randomly initialise K cluster centroids.*

- *Step 2: Assign each data point to its nearest centroid based on distance (typically Euclidean distance).*
- *Step 3: Recompute each centroid as the mean of all points assigned to that cluster.*
- *Step 4: Repeat Steps 2 and 3 until centroids no longer change significantly (convergence).*
- *Example: For K=2 on a set of 2D points, initial centroids are picked randomly; points are assigned to the nearer centroid; centroids are updated to the average position of their assigned points; this repeats until the clusters stabilise into two well-separated groups.*

(b) Choosing K using the Elbow Method:

The model is run for a range of K values (e.g., 1 to 10), and the Within-Cluster Sum of Squares (WCSS) is plotted against K. WCSS decreases as K increases, but after a certain point the rate of decrease slows sharply, forming an "elbow" shape in the graph. The K value at this elbow point is chosen as it gives a good trade-off between cluster compactness and model simplicity.

(c) K-Means vs Hierarchical Clustering:

Aspect	*K-Means*	*Hierarchical Clustering*
Approach	*Partition-based*	*Builds a tree/dendrogram (agglomerative or divisive)*
Need for K	*Must specify K in advance*	*K can be chosen later by cutting the dendrogram*
Scalability	*Efficient for large datasets*	*Computationally expensive for large datasets*
Cluster shape	*Works best with spherical clusters*	*Can capture nested/irregular cluster structures*
Result stability	*Sensitive to initial centroid choice*	*Deterministic given the same linkage method*

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5 Marks)**	**Level 3 – Proficient (4 Marks)**	**Level 2 – Developing (2–3 Marks)**	**Level 1 – Beginning (0–1 Mark)**
Conceptual Understanding	*30*	*Demonstrates clear and complete understanding of the concept*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor or incorrect understanding*
Accuracy of Answer	*20*	*Completely correct answer with no errors*	*Mostly correct with minor errors*	*Partially correct with noticeable errors*	*Incorrect or irrelevant answer*
Application of Knowledge	*20*	*Correctly applies concepts to solve the question/problem*	*Applies concepts with minor mistakes*	*Limited or partially correct application*	*No or incorrect application*
Completeness of Response	*15*	*Covers all required parts of the question*	*Covers most parts with minor omissions*	*Covers only some parts*	*Incomplete or missing response*
Clarity & Presentation	*15*	*Clear, concise, and well-organized answer*	*Understandable with minor issues*	*Somewhat unclear or poorly structured*	*Difficult to understand*